

Question Paper

Exam Date & Time: 25-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. EXAMINATIONS - NOVEMBER 2025

Data Analytics and Visualization [CSS_5113]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply your understanding of data analytics to compare and contrast the key methods-Descriptive, Diagnostic, Predictive, and Prescriptive Analytics-and explain how organizations choose the appropriate method based on project objectives and data types. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Understand"
- B) Explain the concept of Data Segmentation and describe why it is significant in data analytics projects. (3)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- C) List any two advantages of density-based clustering methods and describe why they are useful in data analysis. (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 2) Demonstrate the difference between structured and unstructured data using Python with suitable examples. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Explain the concept of Hypothesis Testing in Data Analytics and interpret the purpose of each step involved in the process. (3)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- C) What are the two main types of interestingness measures. Explain with examples. (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Analyze the concept of Exploratory Data Analysis (EDA) and evaluate its role in identifying patterns, detecting anomalies, and improving decision-making in data analytics projects. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Apply your understanding of statistical inference to explain the concept of a confidence interval and illustrate its practical use in real-world data analysis. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Explain the Gestalt Principle of Similarity and describe how it helps viewers interpret visual information effectively, with a suitable example. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 4) Analyze how the Apriori Algorithm identifies frequent itemsets and association rules, and evaluate its strengths and limitations using an example. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Justify the role of the Visualization Reference Model in designing effective visualizations and justify its application using a Python-based visualization. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Summarize the importance of perception techniques in designing effective and meaningful data visualizations (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 5) Evaluate and compare the effectiveness of graph and tree structures in data visualization, highlighting their strengths, limitations, and suitable use cases. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Evaluate the role of interaction and visualization techniques in enhancing data interpretation. Support your answer with a suitable Python example. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Explain why data structures are important in data visualization and how they support effective visual representation of information. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5113:Data Analytics and Visualization

#	Course Outcome	Description	Marks
1	2	CO 2	16
2	3	CO 3	13
3	4	CO 4	10
4	5	CO 5	11
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5113:Data Analytics and Visualization

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	8	21(42%)
3	Application	7	29(58%)
4	Analysis	0	0(0%)

5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)